Computational Fluid Dynamics
EN 530.767
HW 2

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1. Objective

Consider the viscous Burger's equation:

$$u_t + u_x = \frac{1}{Pe} u_{xx} \text{ on } 0 \le x \le 1$$

Solve this equation using FTCS, 1st-order upwind scheme for convection term, and QUICK scheme. Use grids with different spacing ($\frac{1}{20}, \frac{1}{50}, \frac{1}{100}$). Explore the oscillatory phenomena of solutions and the effects of effective viscosity of these schemes.

2. Methodology

2.1 Numerical schemes

1. Using Forward Euler method for time derivative and central difference for space derivative, the FTCS scheme can be written as

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} = \frac{1}{Pe} \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$$

2. Using first-order upwind scheme for the convection term, the scheme becomes

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{u_i^n - u_{i-1}^n}{\Delta x} = \frac{1}{Pe} \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$$

3. For the QUICK scheme,

$$\phi_e = \frac{6}{8}\phi_P + \frac{3}{8}\phi_E - \frac{6}{8}\phi_W$$

Therefore, the convection term can be represented by

$$u_x = \frac{u_e - u_w}{\Delta x} = \frac{\left(\frac{3}{8}u_{i+1} + \frac{6}{8}u_i - \frac{1}{8}u_{i-1}\right) - (\frac{3}{8}u_i + \frac{6}{8}u_{i-1} - \frac{1}{8}u_{i-2})}{\Delta x}$$
$$= \frac{3u_{i+1} + 3u_i - 7u_{i-1} + u_{i-2}}{8\Delta x}$$

Then the QUICK scheme can be written as

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{3u_{i+1} + 3u_i - 7u_{i-1} + u_{i-2}}{8\Delta x} = \frac{1}{Pe}\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$$

2.2 Select time-step size

For the FTCS scheme, from Van-Neumann stability, the stability condition should be

$$\begin{cases} C^2 \leq 2r, \\ r \leq \frac{1}{2}, r = \frac{\nu \Delta t}{\Delta x^2} \end{cases}$$

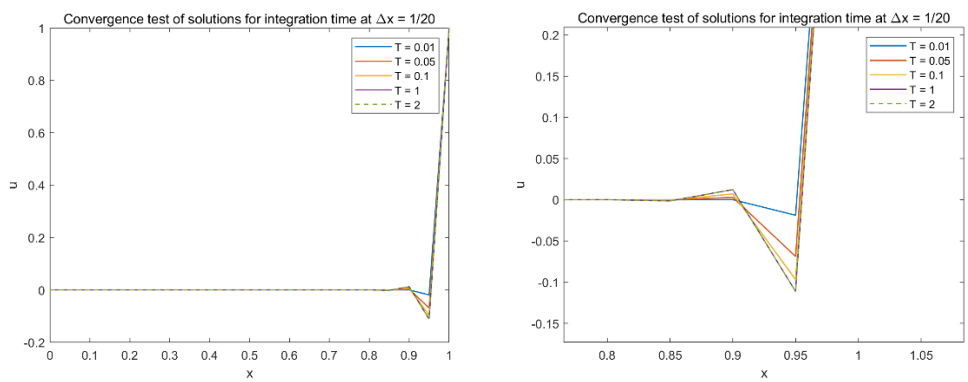
where $C = \frac{c\Delta t}{\Delta x}$ and $r = \frac{\nu \Delta t}{\Delta x^2}$. From the given equation, $c = 1$ and $\nu = 1/50$. For the finest grid, which means $\Delta x = 1/100$, the stability conditions that Δt should satisfy is

$$\begin{cases} \Delta t \leq 0.04 \\ \Delta t \leq 0.0025 \end{cases} \Rightarrow \Delta t \leq 0.0025$$

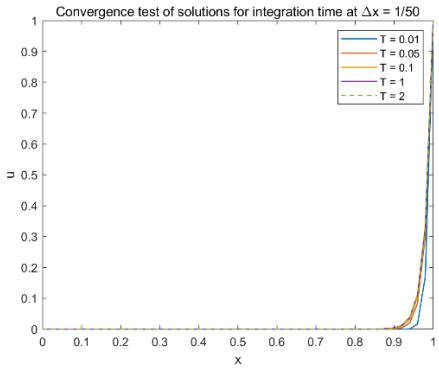
Therefore, we select $\Delta t = 0.0001 \leq 0.0025$ as the time step to solve the equation.

2.3 Convergence examination for integration time

It's apparent that the solution of the Burger's equation is time dependent. Over time, given sufficient integration duration, the solution is expected to converge towards a specific value, aligning with the exact solution at $t = \infty$. To determine this convergent value, we conducted iterative computations for varying lengths of integration time. This approach allowed us to identify the point after which the solution converges to a single value, and we can use the solutions after that point as our numerical solutions to the Burger's equation.



(a) (b)



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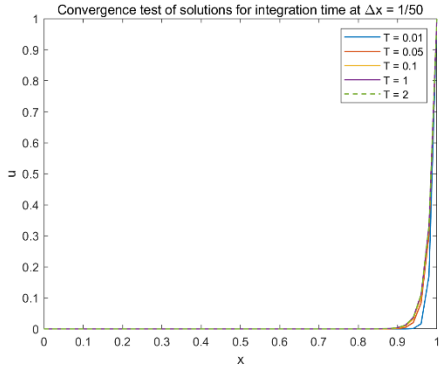

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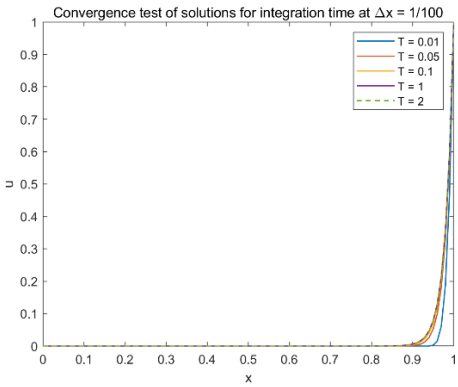
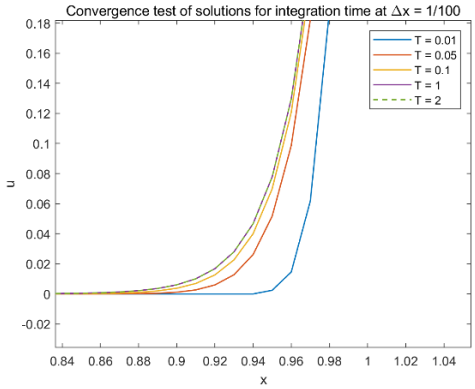


(e)



(f)

Figure 1 presents the numerical solution of the FTCS method with different grid points at various time points. Panel (a), (c), (e)shows the original view, while panel (b), (d), (f) offers a zoomed-in perspective for detailed examination.

Observations from Figure 1 reveal that in the initial time steps, there are noticeable differences between the solutions at various time points. However, beyond $T = 1$, these solutions begin to overlap with each other. Based on this observation, we select the solution at $T = 1$ as the representative final numerical solution for this study.

3. Results and Discussion

3.1 Comparison of solutions solved by different grid sizes

Numerical solutions obtained using grid sizes of varying resolutions with the FTCS scheme, the first-order upwind scheme for the convection term, and the QUICK scheme are collectively displayed in Figures 2 through 4. Within these figures, the numerical solutions corresponding to each grid size are plotted together for direct comparison.

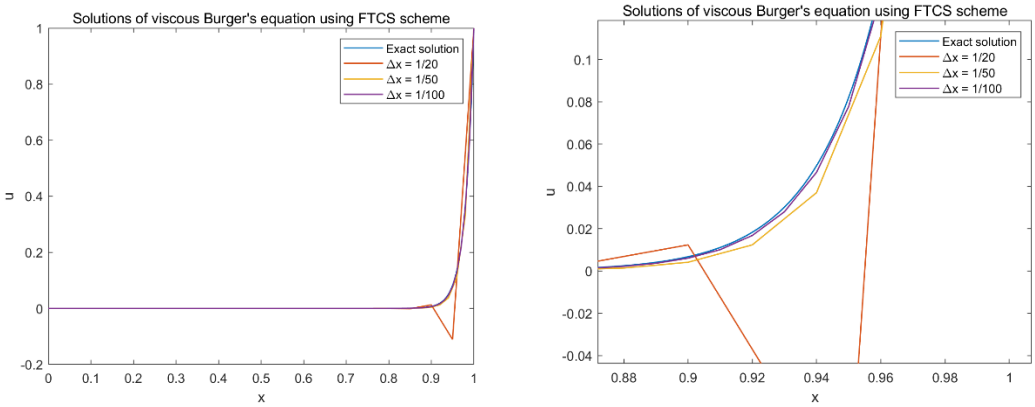
For the FTCS scheme, the solution oscillates at $\Delta x = 1/20$, while the solutions are smooth at $\Delta x = 1/50$ and $\Delta x = 1/100$. This oscillatory phenomenon can be explained theoretically. The FTCS scheme can be written as

$$u_i^{n+1} = \frac{r}{2}(2 - Pe_{\Delta x})u_{i+1}^n + (1 - 2r)u_i^n + \frac{r}{2}(2 + Pe_{\Delta x})u_{i-1}^n$$

For the given initial condition and boundary conditions, we can obtain

$$u_{N-1}^1 = \frac{r}{2}(2 - Pe_{\Delta x})u_N^0 = \frac{r}{2}(2 - Pe_{\Delta x})$$

where $Pe_{\Delta x} = \frac{c\Delta x}{\nu} = 50\Delta x$ in this problem. Therefore, when $Pe_{\Delta x} > 2$, which means $\Delta x > 1/25$, then $u_{N-1}^1 < 0$, and then the solution oscillates. That's why the solution solved by $\Delta x = 1/20$ oscillates, while the solutions solved by $\Delta x = 1/50$ and $\Delta x = 1/100$ are smooth. The behavior of the solutions of the FTCS scheme aligns with the theoretical prediction.



(a) (b)

Figure 2 Numerical solutions solved by different grid sizes using FTCS scheme. Panel (a) shows the original view, while panel (b) offers a zoomed-in perspective for detailed examination.

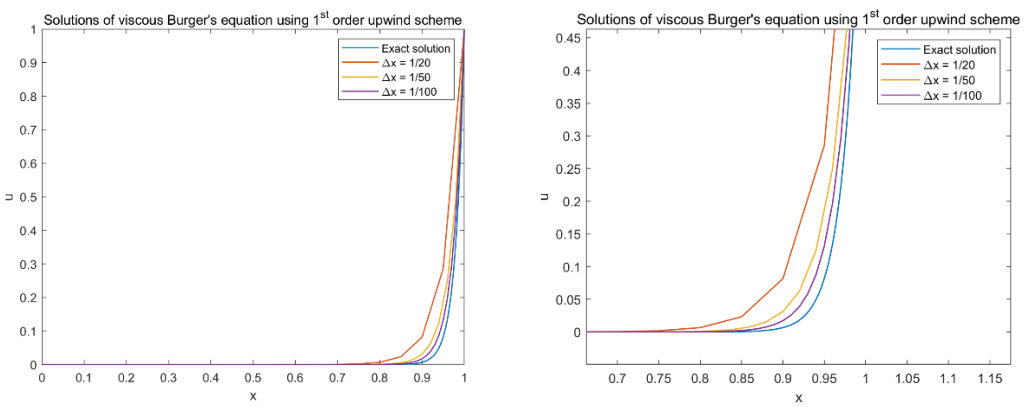


Figure 3 Numerical solutions solved by different grid sizes using 1st-order upwind scheme for convection term. Panel (a) shows the original view, while panel (b) offers a zoomed-in perspective for detailed examination.

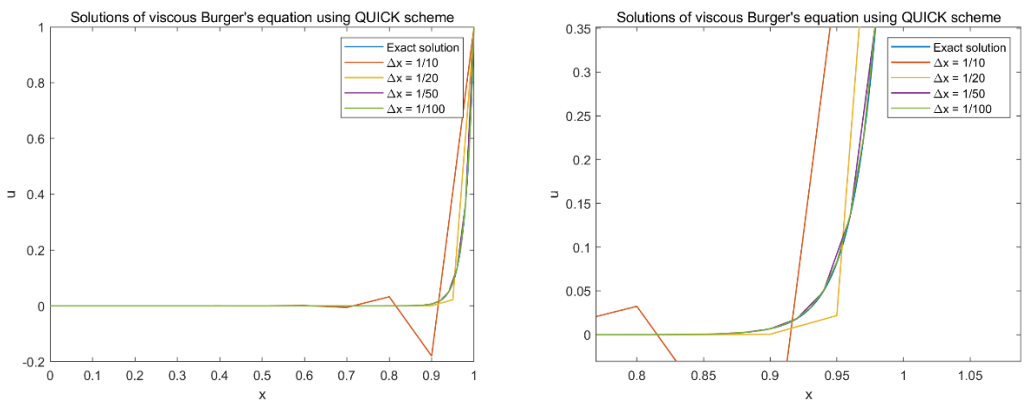


Figure 4 Numerical solutions solved by different grid sizes using QUICK scheme. Panel (a) shows the original view, while panel (b) offers a zoomed-in perspective for detailed examination.

The 1st-order scheme for convection term, instead, will not generate the oscillatory result. The reason can be clarified using the same way. For this scheme,

$$u_i^{n+1} = ru_{i+1}^n + (1 - r(2 + Pe_{\Delta x}))u_i^n + r(1 + Pe_{\Delta x})u_{i-1}^n$$

for which $u_{N-1}^1 = ru_N^0 = r > 0$ for all Δx. Therefore, in Figure 3, we don't see any oscillatory results even the grid size is as coarse as $\Delta x = 1/20$.

The QUICK scheme can be written as

$$u_i^{n+1} = \left(r - \frac{3C}{8}\right)u_{i+1}^n + \left(1 - 2r - \frac{3C}{8}\right)u_i^n + \left(r + \frac{7C}{8}\right)u_{i-1}^n + (-\frac{C}{8})u_{i-2}^n$$

For this case, if $r - \frac{3C}{8} < 0$, which corresponds to $\Delta x > 4/75$, the solution exhibits oscillatory behavior. This is demonstrated in Figure 4 where the solution computed with $\Delta x = 1/10$ is prone to oscillations, in contrast to solutions derived using finer grid resolutions that remain smooth.

Beyond the observed oscillations, Figures 2 through 4 also provide insights into the solution's accuracy. It is evident that utilizing more refined grids leads to numerical solutions that converge more closely towards the exact solution, signaling an improvement in precision. This trend of increased accuracy with finer grids is consistent across all three computational schemes utilized in this analysis.

3.2 Modified equation and effective viscosity

To better understand the behaviors of the numerical solutions, the modified equations of the three schemes are derived (shown in Figure 5). The modified equations are derived up to 2nd-oder spatial derivative.

Modified equation

From Taylor expansion:

$$u_i^{n+1} = u_i^n + u_t \Delta t + \frac{1}{2} u_{tt} \Delta t^2 + \cdots$$

$$u_{i+1}^n = u_i + u_x \Delta x + \frac{1}{2} u_{xx} \Delta x^2 + \cdots$$

$$u_{tt} = (u_t)_t = (\nu u_{xx} - c u_x)_t = \nu (u_t)_{xx} - c (u_t)_x$$

$$= \nu (\nu u_{xx} - c u_x)_{xx} - c (\nu u_{xx} - c u_x)_x$$

$$= c^2 u_{xx} - 2\nu c u_{xxx} + \nu^2 u_{xxxx}$$

For FTCS scheme: $\frac{u_i^{n+1} - u_i^n}{\Delta t} + c \frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} = \nu \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

$$u_t + \frac{\Delta t}{2} u_{tt} + \cdots + c\left(u_x + u_{xxx} \frac{1}{6} \Delta x^2 + \cdots\right) = \nu (u_{xx} + \cdots)$$

Substitute: $u_{tt} = c^2 u_{xx} - 2\nu c u_{xxx} + \nu^2 u_{xxxx}$

$$\Rightarrow u_t + \frac{\Delta t}{2}(c^2 u_{xx} + \cdots) + c(u_x + \cdots) = \nu (u_{xx} + \cdots)$$

$$\Rightarrow u_t + c u_x = \left(\nu - \frac{c^2 \Delta t}{2}\right) u_{xx} + \cdots$$

For 1st order upwind scheme: $\frac{u_i^{n+1} - u_i^n}{\Delta t} + c \frac{u_i^n - u_{i+1}^n}{\Delta x} = \nu \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

$$u_t + \frac{\Delta t}{2} u_{tt} + \cdots + c\left(u_x + \frac{1}{2} u_{xx} \Delta x + \cdots\right) = \nu (u_{xx} + \cdots)$$

$$\Rightarrow u_t + \frac{\Delta t}{2}(c^2 u_{xx} + \cdots) + c\left(u_x - \frac{1}{2} u_{xx} \Delta x + \cdots\right) = \nu (u_{xx} + \cdots)$$

$$\Rightarrow u_t + c u_x = \left(\nu - \frac{c^2 \Delta t}{2} + \frac{c \Delta x}{2}\right) u_{xx} + \cdots$$

$$\Rightarrow u_t + c u_x = \left[\nu \left(1 + \frac{Pe_{\Delta x}}{2}\right) - \frac{c^2 \Delta t}{2}\right] u_{xx} + \cdots \quad \left(Pe_{\Delta x} = \frac{c \Delta x}{\nu}\right)$$

The effective viscosity is $\nu \left(1 + \frac{Pe_{\Delta x}}{2}\right)$

For QUICK scheme: $\frac{u_i^{n+1} - u_i^n}{\Delta t} + c \frac{3u_{i+1} + 3u_i - 7u_{i-1} + u_{i-2}}{8\Delta x} = \nu \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$

$$u_t + \frac{\Delta t}{2} u_{tt} + \cdots + c\left(u_x + \frac{1}{24} u_{xxx} \Delta x^2 + \cdots\right) = \nu (u_{xx} + \cdots)$$

$$\Rightarrow u_t + \frac{\Delta t}{2}(c^2 u_{xx} + \cdots) + c(u_x + \cdots) = \nu (u_{xx} + \cdots)$$

$$\Rightarrow u_t + c u_x = \left(\nu - \frac{c^2 \Delta t}{2}\right) u_{xx} + \cdots$$

Figure 5 Derivation of the modified equations of the three schemes

A higher viscosity implies a higher rate of momentum diffusion within the flow. For both the FTCS and QUICK schemes, the effective numerical viscosity, given by $\nu - \frac{c^2 \Delta t}{2}$, is less than the actual physical viscosity ν. Consequently, the numerical diffusion is expected to be less pronounced than that of the exact solution. This is corroborated by the profiles in Figure 2 and Figure 4, where the exact solutions exhibit sharper gradients compared to the smoother transitions of the numerical solutions, which is in accordance with the theoretical predictions.

For the first-order upwind scheme applied to the convection term, the effective numerical viscosity is $\nu(1 + \frac{Pe_{\Delta x}}{2}) - \frac{c^2 \Delta t}{2}$, thus depending on the grid size. When $\nu\left(1 + \frac{Pe_{\Delta x}}{2}\right) - \frac{c^2 \Delta t}{2} > \nu$, which occurs if $\Delta x > \Delta t$, the effective numerical viscosity

exceeds the actual physical viscosity. In consequence, for numerical solutions computed with $\Delta x = \frac{1}{20}, \frac{1}{50}, \frac{1}{100}$ and $\Delta t = 0.0001$, the solutions are expected to be steeper. This is indeed the case in Figure 3, where the numerical solutions exhibit sharper transitions, confirming the theoretical expectations.

3.3 Comparison of solutions solved by different schemes

In order to investigate the accuracy of different schemes, numerical solutions obtained using grid size equal to $\Delta x = 1/100$ with the FTCS scheme, the first-order upwind scheme for the convection term, and the QUICK scheme are collectively displayed again in Figure 6. For this time, the numerical solutions corresponding to each scheme are plotted together for direct comparison.

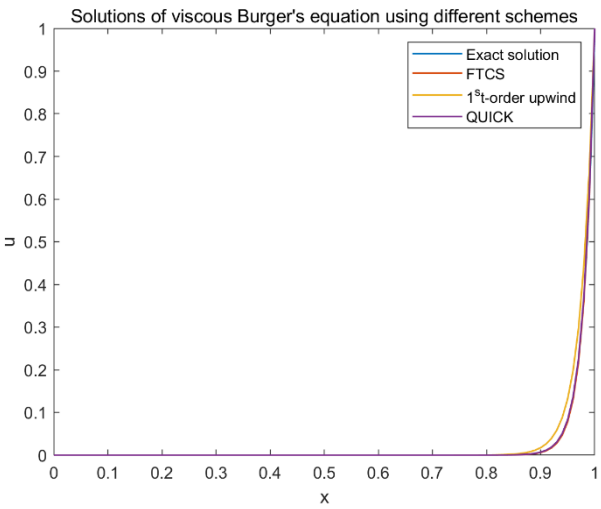


Figure 6 Numerical solutions solved by different schemes at $\Delta x = 1/100$.

From Figure 6, it is observed that the numerical solutions obtained from the FTCS and QUICK schemes closely approximate the exact solution. In contrast, the solution computed using the first-order upwind scheme for the convection term exhibits some deviation. This discrepancy can be attributed to the fact that in the discretization of the convection term, the FTCS scheme is second-order accurate, and QUICK possesses a higher-order accuracy, while the first-order upwind scheme, as the name suggests, only has first-order accuracy. This aligns with our numerical findings.

4. Conclusion

In conclusion, our computational investigation into the viscous Burger's equation has demonstrated distinct characteristics of numerical solutions when applied with FTCS, first-order upwind, and QUICK schemes across various grid resolutions. The observed oscillatory behavior in coarser grids and the effect of effective viscosity further underline the importance of scheme selection in computational fluid dynamics simulations. The study confirms that finer grids enhance solution accuracy, a consistent trend across all tested numerical schemes. Our findings also reveal that

while FTCS and QUICK schemes deliver numerical solutions that closely match the exact solution, the first-order upwind scheme displays a slight divergence due to its lower order of accuracy. This aligns with theoretical expectations, as the FTCS and QUICK schemes incorporate higher-order discretization in the convection term, providing more accurate results.

5. Appendix

MATLAB source code:

```
clear; clc;
Pe = 50;
dt = 0.0001;
T = 10;
scheme = 1;

%% Convergence test
dx = 1/20;
figure
[xn, un] = FTCS(dt, dx, Pe, 0.01, scheme);
plot(xn, un, 'linewidth', 1)
hold on
[xn, un] = FTCS(dt, dx, Pe, 0.05, scheme);
plot(xn, un, 'linewidth', 1)
hold on
[xn, un] = FTCS(dt, dx, Pe, 0.1, scheme);
plot(xn, un, 'linewidth', 1)
hold on
[xn, un] = FTCS(dt, dx, Pe, 1, scheme);
plot(xn, un, 'linewidth', 1)
hold on
[xn, un] = FTCS(dt, dx, Pe, 2, scheme);
plot(xn, un, '--', 'linewidth', 1)

title('Convergence test of solutions for integration time at \Deltax =
1/20')
xlabel('x')
ylabel('u')
legend('T = 0.01', 'T = 0.05', 'T = 0.1', 'T = 1', 'T = 2')

%% Plot solutions 1
x = linspace(0, 1, 1000);
uexa = (exp(Pe.*x)-1)./(exp(Pe)-1);
figure;
```

```
plot(x, uexa, 'linewidth', 1)

dx = 1/10;
[xn, un] = FTCS(dt, dx, Pe, T, scheme);
hold on
plot(xn, un, 'linewidth', 1)

dx = 1/20;
[xn, un] = FTCS(dt, dx, Pe, T, scheme);
hold on
plot(xn, un, 'linewidth', 1)

dx = 1/50;
[xn, un] = FTCS(dt, dx, Pe, T, scheme);
hold on
plot(xn, un, 'linewidth', 1)

dx = 1/100;
[xn, un] = FTCS(dt, dx, Pe, T, scheme);
hold on
plot(xn, un, 'linewidth', 1)

title('Solutions of viscous Burger''s equation using QUICK scheme')
xlabel('x')
ylabel('u')
legend('Exact solution', '\Deltax = 1/10', '\Deltax = 1/20', '\Deltax =
1/50', '\Deltax = 1/100')

%% Plot solutions 2
x = linspace(0, 1, 1000);
uexa = (exp(Pe.*x)-1)./(exp(Pe)-1);
dx = 1/100;
figure;
plot(x, uexa, 'linewidth', 1)

scheme = 1;
[xn, un] = FTCS(dt, dx, Pe, T, scheme);
hold on
plot(xn, un, 'linewidth', 1)

scheme = 2;
[xn, un] = FTCS(dt, dx, Pe, T, scheme);
hold on
plot(xn, un, 'linewidth', 1)
```

```
scheme = 3;
[xn, un] = FTCS(dt, dx, Pe, T, scheme);
hold on
plot(xn, un, 'linewidth', 1)

title('Solutions of viscous Burger''s equation using different schemes')
xlabel('x')
ylabel('u')
legend('Exact solution', 'FTCS', '1^st-order upwind', 'QUICK')

%% Functions
function [xn, un] = FTCS(dt, dx, Pe, T, scheme)
N=floor(1/dx);
xn = linspace(0, 1, N+1);
n = T/dt;

un=zeros(N+1,1);
un(N+1) = 1;
unew = un;
C = dt/dx;
r = dt/Pe/dx^2;

for i = 1:n
    for j = 2:N
        if scheme == 1
            unew(j) = (r-C/2)*un(j+1) + (1-2*r)*un(j) + (r+C/2)*un(j-1);
        end
        if scheme == 2
            unew(j) = r*un(j+1) + (1-C-2*r)*un(j) + (r+C)*un(j-1);
        end

    end
    un = unew;
end

if scheme == 3
    ug = 2*un(1)-un(2);
    for i = 1:n
        j=2;
        unew(j) = (r-3*C)*un(j+1) + (1-3*C-2*r)*un(j) + (r-5*C)*un(j-1) +
C*ug;
        for j = 3:N
```

```
            unew(j) = (r-3*C/8)*un(j+1) + (1-3*C/8-2*r)*un(j) +
(r+7*C/8)*un(j-1) + (-C/8)*un(j-2);
        end
        un = unew;
        ug = 2*un(1)-un(2);
    end
end
un = un';

end
```